

Documentation File (Dossier)

- **App Name:** AACARDS
- **Module:** Introduction to App Development
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- **Date:** 12.01.2026
- **Lecturer:** Mr.Prof.Heller



AACARDS

1.Problem the App Solves

Students and pupils often face uncertainty about how well they understand individual topics and how effective their exam preparation is. The goal of this app is to help students independently test themselves by creating flashcards for each module, breaking down topics into detailed subtopics, and writing summaries.

According to numerous studies, this method—*active recall*—is one of the most effective ways to learn. The app supports this learning method by allowing teachers to actively evaluate students' summaries and create quizzes, helping them reinforce what they have learned. Additionally, students can upload files to individual flashcards, and teachers can correct them and re-upload the corrected files for students to download.

The app aims to achieve the following:

- Students and pupils achieve better grades.
- Students gain a deep understanding of each topic. Detailed attention is given to each subtopic (which will be explained further below).
- Students improve their self-directed learning skills.
- Teachers can monitor students' independent learning and support them from the beginning.

2.Target Audience

- Pupils from grade 5 onwards
- University students
- School lecturers / teachers
- University lecturers / professors

3.Why the Problem is Relevant

I once submitted a modeling task to a professor, who provided detailed analysis and feedback. I found this extremely effective because I could see exactly where my knowledge gaps were, why I made certain mistakes, and which concepts I had fully understood. Compared to standard solution keys, this method was far more effective.

If such detailed feedback is available for every single aspect of a topic—as a flashcard—this creates very high value for learning.

4.App Idea & Use Cases

- **As a Student/Pupil:**
 - I want an app where I can create, edit, and delete flashcards for my learning process, so I can record key points and use them for exam preparation.
 - I want an app where teachers can monitor my learning progress and provide active feedback.
- **As a Teacher:**
 - I want an app where I can track the learning progress of my students or pupils, correct their notes, download and review their uploaded assignments, and re-upload the corrected versions.
 - I also want to create quizzes to test my students or pupils.

5.Main Features

- Login / Registration
- Select Students
- Create, edit, delete flashcards
- Write summaries (Student)
- Provide feedback (Teacher)
- Upload / Download files (Teacher / Student)
- Create quizzes (Teacher)
- Take quizzes (Student)

6.Mockups / Sketches (Paper Prototype)

Name

Password

Login

Register

Willkommen bei Study APP

Login

Willkommen Abdul

~~Willkommen XY~~

! Login für Student / holt was
Dozent angelegt wurden

Studiengang willkommen ~~Abdul~~
wählen

Math1

Physik

APP Dev

Register

Uni wählen ~~Register~~ ~~Abdul~~ ~~Abdul~~
Name

Passwort
~~Neutiger~~
~~Modul~~ ~~entfernen~~

~~Stud. auf wählen~~

~~Modul~~ ~~entfernen~~

Math1

Math1

~~APPTHEM~~

⋮ Thema

Math 1 / Algebra

Arithmetik

~~ADD button~~

Fortfahren

⋮ Arithmetik

Math 1 / Algebra / Arithmetik

Karten hinzufügen

Quiz

besten
herunterladen:

V1
Ü1

Ü1

V1

Feedback received:

zu Ü1
zu V1

Fortfahren

⋮ Karten

title

Notiz

Feedback

↓ Thema

Math 1 / Algebra

Arithmetik

~~add button~~

Fortfahren

↓ Arithmetik

Math 1 / Algebra / Arithmetik

Karten hinzufügen

Quiz

~~button~~
vermischen:
V1
Ü1

Ü1

V1

Feedback received:

zu Ü1
zu V1

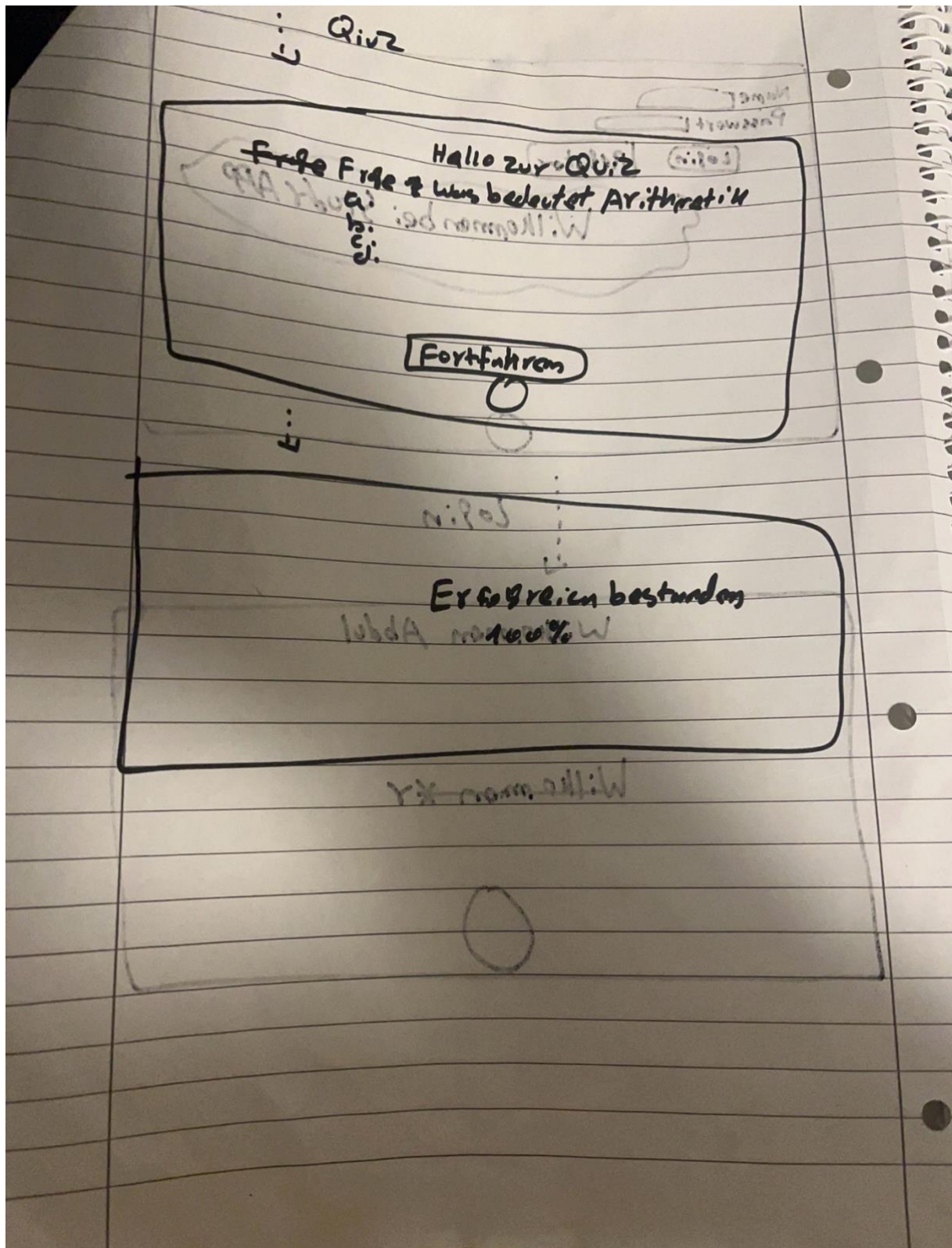
Fortfahren

↓ Karten

title

Notiz

Feedback

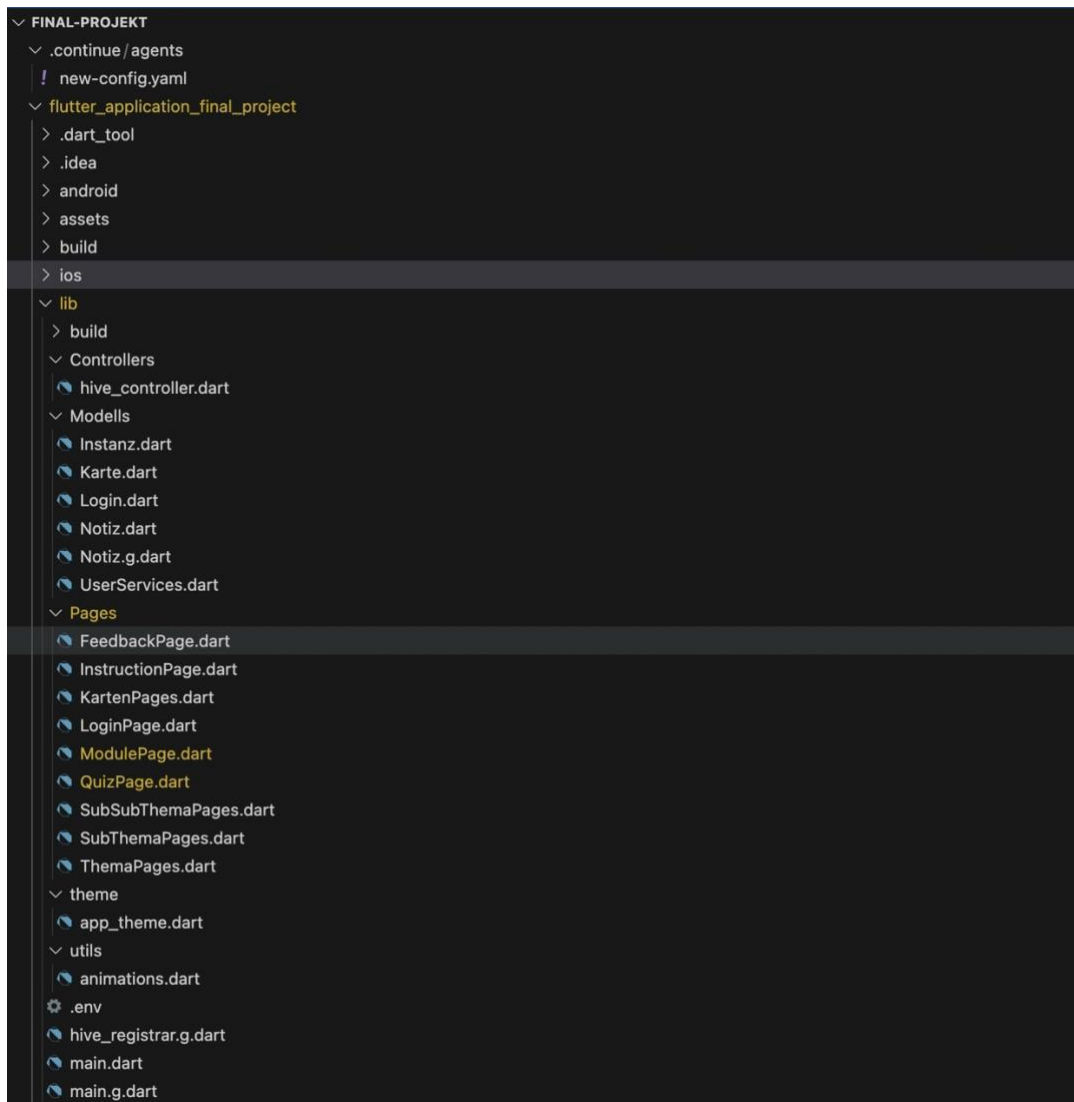


7. Implementation

7.1 Architecture

The app follows a **Client-Server Architecture**.

The app has been developed using the Flutter framework (Dart language) in Visual Studio Code (VS Code).



Frontend:

- **QuizPage:**
 - When a student clicks on "Start Quiz," they can take a quiz if available.
 - When a lecturer clicks on "Create Quiz," they can generate a new quiz. Before that, an instruction page is displayed for the lecturer explaining how to create a quiz.
- **ModulePage:**
 - This is the welcome page after login/registration.
 - It displays the profile and allows users to create modules for their program.
- **FeedbackPage:**
 - Students can view feedback if available.
 - Lecturers can write feedback.
- **ThemaPages (Topic Page):**
 - After clicking on a module, the user is directed here.
 - They can create cards for this module.

- **SubThemaPages (Subtopic Page):**
 - After clicking on a Thema-Card, the user is directed here.
 - They can create cards for this Topic.
- **SubSubThemaPages (Sub-Subtopic Page):**
 - After clicking on a SubThema-Card, the user is directed here.
 - They can create cards for this subtopic.
- **LoginPage:**
 - This is the first page of the app when it starts.
 - Users must Login to access their account.
- **InstructionPage:**
 - After Login, instructions are shown depending on the user's role, explaining how to use the app.
- **KartenPages:**
 - Users can write summaries, upload/download files, give feedback, start quizzes, create quizzes, and write feedback for a subsubtopic.
- **app_theme:**
 - Custom colors, fonts, and overall app design.
- **animations:**
 - Part of the app design for smooth transitions and visual effects.

Backend:

Supabase Database:

Database Tables:

- **Universities → programs → modules → Theme → SubTheme → SubSubTheme → Karte → Quiz**
- Each table connected with a foreign key to the previous one.
 - Example: Program has a foreign key referencing Uni.
- **Student**
- **Lecturer**

Data Access:

- Lecturers see relevant students based on the Program ID (same university and program).
- When a lecturer selects a student, the correct student data is loaded based on the student_Id in the Table „Module“.

Authentication:

- **Users Table** manages login and registration.

Storage:

- Used for file upload and download.
 - Example: space-cat.pdf → CardID → FileName

The database credentials are stored securely in a .env file.

The UserService class, located in the Modells folder, handles all operations related to database access and user data management.

7.2 Technologies Used / Tools:

Dependencies (App-Pakete)

- flutter (SDK)
- cupertino_icons
- supabase_flutter
- envied
- uuid
- hive_ce
- hive_ce_flutter
- hive_ce_generator
- crypto
- lib
- flutter_login
- image_picker
- file_picker
- file_save_directory
- quick_quiz
- introduction_screen
- animations

Dev Dependencies (Entwicklungs-/Build-Pakete)

- **flutter_test** (SDK)
- **flutter_lints**
- **envied_generator**
- **build_runner**
- **flutter_launcher_icons**

8.User Flow / Navigation

LoginPage → InstructionPage → ModulePage → ThemaPages → SubThemaPages
→ SubSubThemaPages → KartenPages → QuizPage/FeedbackPage

9.Limitations / Assumptions

the app is developed with Flutter and can theoretically run on all major platforms. The app currently assumes a stable internet connection and does not provide an offline mode, Offline support **could** be implemented using local storage (e.g., Hive), as we did before for caching data and enabling offline functionality.

10.Conclusion / Summary:

In conclusion, the app promotes self-directed learning by replacing traditional paper-based flashcards with a well-structured digital solution.

Through its modular architecture, students can systematically organize their learning content, while teachers are able to evaluate summaries and provide targeted feedback.

Additional features such as quizzes and file uploads further strengthen knowledge retention and support active recall as an effective learning method.

Overall, the app demonstrates how digital tools can enhance learning processes in a meaningful and scalable way.

11.References / Bibliography (optional):

Lecture slides and materials from the course *Introduction to App Development*

- Flutter Documentation, <https://flutter.dev>
- Supabase Documentation, <https://supabase.com>

12.Fotos from the App:

Note: Abdulrahman is a Student and Nabil is a Lecturer

